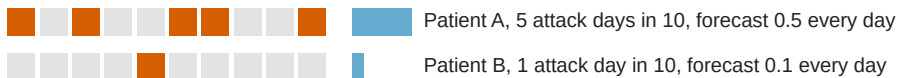


(a) Worked cohort: a classifier that outputs each patient's own attack rate, and nothing else



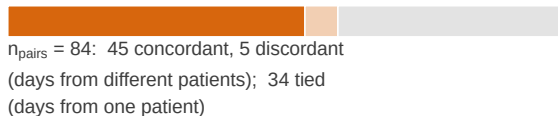
The forecast never varies across a patient's days, so it carries no day-level information whatever. Read three ways:

(b) Discrimination: AUROC counts ordered pairs

The chance a randomly drawn attack day is ranked above a randomly drawn quiet day, ties counting one half:

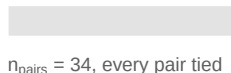
$$\text{AUROC} = \frac{n_{\text{concordant}} + \frac{1}{2}n_{\text{tied}}}{n_{\text{pairs}}}$$

(c) Pooled: pairs from the whole cohort



$$\text{AUROC} = \frac{45 + \frac{1}{2}(34)}{84} = 0.74$$

(d) Within person: pairs inside each patient



$$C = \frac{0 + \frac{1}{2}(34)}{34} = 0.50$$

(e) Clinical value: Brier skill

Squared error of the forecast, set against simply using each patient's own attack rate. Zero means the forecast adds nothing to that rate:

$$\text{skill} = 1 - \frac{\text{Brier}(\text{forecast})}{\text{Brier}(\text{own rate})}$$

(f) The same classifier, scored this way



Identical, because the classifier is that rate: it reproduces the comparator exactly.

$$\text{skill} = 1 - \frac{0.17}{0.17} = 0.00$$

One classifier, three readings: 0.74 pooled, 0.50 within a patient, 0.00 against the patient's own rate. Only the first suggests a usable forecast.